Higher order differential equation

- Homogeneous linear DEs

$ay'' + by' + cy = 0 \qquad y(x)\ ?$

$y(x) = e^{mx}$ m is constant

$a(m^2 \cdot e^{mx}) + b(me^{mx}) + c(e^{mx}) = 0$

$e^{mx}[am^2 + bm + c] = 0$ — e^{mx} is always going non-zero.

roots m_1 & m_2

The auxiliary equation is $am^2 + bm + c = 0$

- The 2 roots: $m_1 = \dfrac{-b + \sqrt{b^2 - 4ac}}{2a} \qquad m_2 = \dfrac{-b + \sqrt{b^2 - 4ac}}{2a}$

$\rightarrow y_2(x) = e^{m_2 x}$

Case 1: m_1 and m_2 we real and distinct $y = C_1 e^{m_1 x} + C_2 e^{m_2 x}$

$\rightarrow y_1(x) = e^{m_1 x}$

Case 2: m_1 and m_2 are real and equal

$y_1(x) = e^{m_1 x}$

$y_2(x) = \underbrace{\boxed{\uparrow}}_{u(x)} y_1(x) \rightarrow \quad y_2(x) = y_1(x) \int \dfrac{e^{-\int P(x)\,dx}}{y_1^2(x)}$

$ay'' + by' + cy = 0 \rightarrow y'' + \underbrace{\frac{b}{a}}_{P(x)} y' + \frac{c}{a} y = 0$

$m_1 = m_2 = \dfrac{-b}{2a} \rightarrow \quad P(x) = 2m_1$

Ex 1: Solve $2y'' - 5y' - 3y = 0$

aux eqn: $2m^2 - 5m - 3 = 0 \rightarrow -\frac{b \pm \sqrt{b^2-4ac}}{2a}$

or $\rightarrow (2m+1)(m-3) = 0 \Rightarrow m_1 = 3,\ m_2 = -\frac{1}{2}$

$y(x) = c_1 e^{3x} + c_2 e^{-\frac{x}{2}}$ Solution

E2: Solve $y'' - 10y' + 25y = 0$

aux eqn: $m^2 - 10m + 25 = 0 \quad (m-5)^2 = 0 \quad m_1 = m_2 = 5$ Case 2

$y(x) = c_1 e^{5x} + c_2 x e^{5x}$

α = Alpha
β = Beta

E3: $y'' + 4y' + 7y = 0$ Case 3

aux eqn: $m^2 + 4m + 7 = 0 \rightarrow \frac{-4 \pm \sqrt{16-28}}{2} = \underset{\alpha}{-2} \pm \underset{\beta}{i\sqrt{3}}$

$-2 + i\sqrt{3} = m_1$

$-2 - i\sqrt{3} = m_2$

$\alpha = -2 \quad \beta = \sqrt{3}$

$y(x) = e^{-2x}(c_1 \cos\sqrt{3}x + c_2 \sin\sqrt{3}x)$

Ex 4: Solve the IVP $4y'' + 4y' + 17y = 0 \quad y(0) = -1,\ y'(0) = 2$

aux eqn: $4m^2 + 4m + 17 = 0 \rightarrow m_1 = \frac{-4 \pm \sqrt{16-272}}{8} = -\frac{1}{2} + 2i = m_1$

$= -\frac{1}{2} - 2i = m_2$

$\alpha = -\frac{1}{2} \quad \beta = 2 \Rightarrow y(x) = e^{-\frac{x}{2}}(c_1 \cos 2x + c_2 \sin 2x)$

$y(0)=1$

$-1 = 1(C_1 \cdot 1 + C_2 \cdot 0) \Rightarrow C_1 = -1$

$y'(0)=2$ he didn't show the process

$C_2 = \frac{3}{4} \Rightarrow y(x) = e^{-\frac{x}{2}}(-\cos x + \frac{3}{4}\sin 2x)$

Higher order Equations

Ex5: Solve $y''' + 3y'' - 4y = 0$ 3rd order

aux equ: $m^3 + 3m^2 - 4 = 0 \Rightarrow (m-1)(m^2+4m+4) = 0$ long division

Case 1 $(m+2)^2$ Case 2

$m_1 = 1,\ m_2 = -2x,\ m_3 = -2$

$$y(x) = C_1 e^{1x} + C_2 e^{-2x} + C_3 x e^{-2x}$$

Ex6: Solve $y'''' + 2y'' + y = 0$

aux equ: $m^4 + 2m^2 + 1 = 0 \rightarrow (m^2+1)^2 = 0$ $\quad m_1 = m_2 = i$, $m_3 = m_4 = -i$

$$y = C_1 e^{ix} + C_2 x e^{ix} + C_3 e^{-ix} + C_4 x e^{-ix}$$

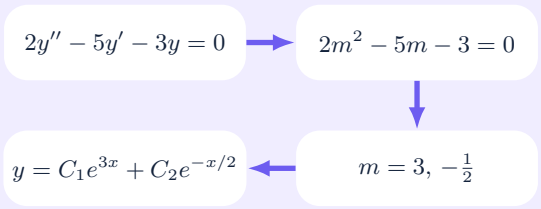
Euler's formula: $y = C_1 \cos x + C_2 \sin x + x(C_3 \cos x + C_4 \sin x)$

MADE FROM YOUR HANDWRITTEN PAGES + CLASS RECORDING

Understand the class. Not just the answers.

Higher-order differential equations, made visual and easy to follow.

TODAY'S ENTIRE METHOD IN ONE GLANCE



Your learning route

01 Understand what equation you're looking at

02 Discover why derivatives become powers of m

03 Learn the three root cases without guessing

04 Solve all six actual class examples with missing steps filled in

05 Use the final checklists to recognize exam questions quickly

Your notes come FIRST in this PDF. Those three pages are preserved exactly as provided. This illustrated explanation starts immediately after them. Your instructor's slides are not reproduced.

Extra reasoning and intermediate calculations in the following pages were added for learning; the examples themselves come from the October 21 class.

01 / SPOT THE PATTERN

What kind of ODE is this?

Before solving anything, check these three properties.

THE MAIN TEMPLATE

$$\boxed{ay'' + by' + cy = 0}$$

LINEAR

No y^2, no $(y')^2$, and no $\sin(y)$.

HOMOGENEOUS

The entire right side is 0.

CONSTANT

The numbers a, b, c do not depend on x.

Quick test

$2y'' - 5y' - 3y = 0$

$y'' + xy' + y = 0$

$y'' + (y')^2 + y = 0$

$y'' + 2y = \cos x$

YES: this lecture

NO: x varies

NO: nonlinear

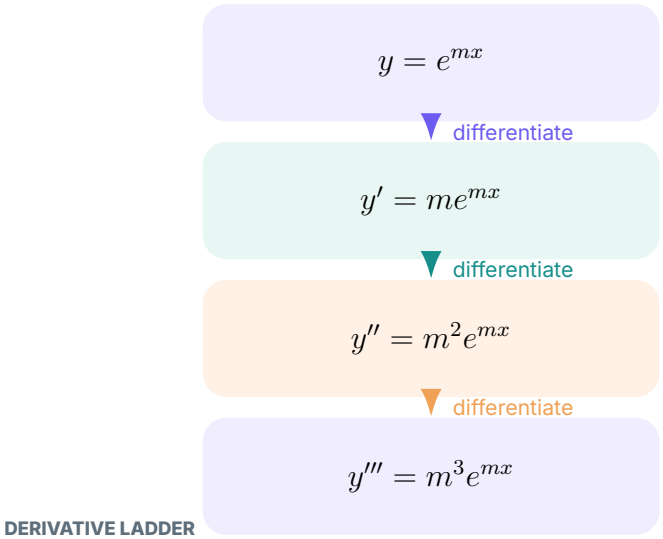
NO: not homogeneous

Don't mix this up: Earlier, "homogeneous" sometimes meant using $v = y/x$ for a first-order ODE. That's a *different* method. Today, right-hand side $= 0$ is the key.

02 / THE TRICK BEHIND IT

Why use e^{mx}?

Your professor chose an exponential because it keeps its shape when differentiated.



DERIVATIVE LADDER

The exam shortcut

y	y'	y''	y'''	y''''
1	m	m^2	m^3	m^4

Why does a new m appear? The chain rule says $\frac{d}{dx}e^{mx} = e^{mx}\frac{d}{dx}(mx) = me^{mx}$. Differentiating once more multiplies by m again.

03 / THE MAIN DERIVATION

Meet the auxiliary equation

This turns a difficult-looking ODE into ordinary algebra.

STEP 1 **Start with a second-order ODE**

$$ay'' + by' + cy = 0$$

STEP 2 **Try $y = e^{mx}$ and substitute the derivatives**

$$a(m^2e^{mx}) + b(me^{mx}) + c(e^{mx}) = 0$$

STEP 3 **Factor the common e^{mx}**

$$e^{mx}\left(am^2 + bm + c\right) = 0$$

STEP 4 **The exponential is NEVER zero. So the bracket must be zero.**

$$\boxed{am^2 + bm + c = 0}$$

This is the auxiliary equation. You can also call it the *characteristic equation*. Find its roots first; those roots tell you what the general solution looks like.

Common confusion: We are not claiming $e^{mx} = 1$. We're dividing it out because it cannot be zero. That's why the exponential disappears from the polynomial.

04 / THE ROOT DECODER

Three cases. Three formulas.

Learn to recognize the type of roots before writing any solution.

1 **Two DIFFERENT real roots** like 3 and -2

$$\boxed{y = C_1 e^{m_1 x} + C_2 e^{m_2 x}}$$

2 **Two EQUAL real roots** like 5 and 5

$$\boxed{y = C_1 e^{mx} + C_2 x e^{mx}}$$

3 **Two COMPLEX roots** like $-2 \pm 3i$

$$\boxed{y = e^{\alpha x}(C_1 \cos \beta x + C_2 \sin \beta x)}$$

How to choose instantly: the discriminant

$$\Delta = b^2 - 4ac$$

$\Delta > 0$	$\Delta = 0$	$\Delta < 0$
Case 1	**Case 2**	**Case 3**
Different	Repeated	Complex

For non-monic quadratics, remember $m = (-b \pm \sqrt{b^2 - 4ac})/(2a)$.

EXAMPLE 01 / CASE 1

Different roots = two exponentials

A complete walkthrough of the first problem from your notes.

QUESTION

$$\boxed{2y'' - 5y' - 3y = 0}$$

1 Replace the derivatives with powers of m

$$2m^2 - 5m - 3 = 0$$

2 Factor. Here's every small step.

$$\begin{aligned} 2m^2 - 5m - 3 &= 0 \\ 2m^2 - 6m + m - 3 &= 0 \\ 2m(m-3) + 1(m-3) &= 0 \\ (2m+1)(m-3) &= 0 \end{aligned}$$

3 Set each bracket equal to zero

$$2m + 1 = 0 \Rightarrow m = -\tfrac{1}{2} \qquad m - 3 = 0 \Rightarrow m = 3$$

4 Different real roots, so it's Case 1.

Just place each root in its own exponential.

FINAL ANSWER

$$\boxed{y = C_1 e^{3x} + C_2 e^{-x/2}}$$

Where did -6 and $+1$ come from? They multiply to $2(-3) = -6$ and add up to -5. Splitting the middle term lets you factor by grouping. You can always use the quadratic formula instead.

EXAMPLE 02 / CASE 2

Repeated roots: don't forget x

This is why the professor kept emphasizing the extra x.

QUESTION

$$\boxed{y'' - 10y' + 25y = 0}$$

1 Auxiliary equation

$$m^2 - 10m + 25 = 0$$

2 Factor: the same bracket appears twice

$$(m-5)(m-5) = 0 \Rightarrow (m-5)^2 = 0$$

3 Both roots are 5

$$m_1 = m_2 = 5$$

4 Repeated real root = Case 2. Add x to term two.

FINAL ANSWER

$$\boxed{y = C_1 e^{5x} + C_2 x e^{5x}}$$

Why not $C_1 e^{5x} + C_2 e^{5x}$? Because that simplifies to $(C_1 + C_2)e^{5x}$, just one independent solution. An order-2 equation needs two independent solutions.

Optional explanation from the lecturer's reduction-of-order discussion: Starting from $y_1 = e^{mx}$, a second solution can be shown to become xe^{mx} when the characteristic root repeats. For solving these problems, learn the final rule.

05 / UNDERSTAND THE EXTRA x

What does “independent” mean?

This connects directly to your previous class on the fundamental set.

A SECOND-ORDER ODE REQUIRES TWO GENUINELY DIFFERENT SOLUTIONS

NOT enough

e^{5x} and $4e^{5x}$

The second is just 4 times the first. They describe the same direction.

Enough

e^{5x} and xe^{5x}

The extra x makes the second one genuinely different.

What your handwritten reduction-of-order notes are doing
The notes show a general formula for finding a second solution:

$$y_2 = y_1 \int \frac{e^{-\int P(x)\,dx}}{y_1^2}\,dx$$

When the roots repeat, $m = -b/(2a)$ and $P(x) = b/a = -2m$. With $y_1 = e^{mx}$, the fraction becomes:

$$\frac{e^{2mx}}{(e^{mx})^2} = 1 \quad \Rightarrow \quad \int 1\,dx = x$$

That gives $y_2 = xe^{mx}$. **This derivation is to understand, not redo on every exam problem.**

Mini memory rule: For repeated roots, the first solution has no extra x. The next one gets x. If the same root appeared three times, the third would get x^2.

EXAMPLE 03 / CASE 3

Complex roots without panic

The imaginary number i eventually turns into sine and cosine.

QUESTION

$$\boxed{y'' + 4y' + 7y = 0}$$

1 Write the auxiliary equation

$$m^2 + 4m + 7 = 0$$

2 Use the quadratic formula

$$\begin{aligned} m &= \frac{-4 \pm \sqrt{4^2 - 4(1)(7)}}{2(1)} \\ &= \frac{-4 \pm \sqrt{16 - 28}}{2} \\ &= \frac{-4 \pm \sqrt{-12}}{2} \end{aligned}$$

3 Simplify the negative radical

$$\sqrt{-12} = \sqrt{4 \cdot 3 \cdot (-1)} = 2i\sqrt{3}$$

$$m = \frac{-4 \pm 2i\sqrt{3}}{2} = -2 \pm i\sqrt{3}$$

4 Separate the two numbers

$$\underbrace{-2}_{\alpha} \quad \pm \quad i\underbrace{\sqrt{3}}_{\beta}$$

FINAL ANSWER

$$\boxed{y = e^{-2x}(C_1 \cos(\sqrt{3}x) + C_2 \sin(\sqrt{3}x))}$$

The mapping: $\alpha = -2$ becomes the exponential e^{-2x}. $\beta = \sqrt{3}$ goes inside $\cos(\sqrt{3}x)$ and $\sin(\sqrt{3}x)$. Do not put i in the final real answer.

06 / WHY TRIG APPEARS

How can i become cosine?

A tiny visual explanation of Euler's formula from class.

EULER'S FORMULA

$$\boxed{e^{i\theta} = \cos\theta + i\sin\theta}$$

START **Take one complex root** $m = \alpha + i\beta$

$$e^{mx} = e^{(\alpha+i\beta)x}$$

SPLIT **Separate the real and imaginary exponentials**

$$e^{\alpha x}e^{i\beta x}$$

EULER **Convert the imaginary exponential**

$$e^{\alpha x}(\cos\beta x + i\sin\beta x)$$

COMBINE **Using the conjugate root, build two real solutions**

$$e^{\alpha x}\cos\beta x, \qquad e^{\alpha x}\sin\beta x$$

You don't need complex exponentials in your final real-valued solution. The professor showed the complex form is possible, but the sine-and-cosine form is normally cleaner.

Try this mentally

Root $m = -3 \pm 2i$. Put -3 in the exponential and 2 inside trig:

$$\boxed{y = e^{-3x}(C_1\cos 2x + C_2\sin 2x)}$$

EXAMPLE 04 / INITIAL VALUES

Same method. Two extra conditions.

The class's IVP had a missing derivative calculation. Here it is in full.

QUESTION FROM CLASS

$$\boxed{4y'' + 4y' + 17y = 0}$$

$$y(0) = -1, \qquad y'(0) = 2$$

1 Auxiliary equation

$$4m^2 + 4m + 17 = 0$$

2 Quadratic formula

$$\begin{aligned} m &= \frac{-4 \pm \sqrt{4^2 - 4(4)(17)}}{2(4)} \\ &= \frac{-4 \pm \sqrt{16 - 272}}{8} \\ &= \frac{-4 \pm \sqrt{-256}}{8} \\ &= \frac{-4 \pm 16i}{8} = -\tfrac{1}{2} \pm 2i \end{aligned}$$

3 Complex roots: $\alpha = -\frac{1}{2},\ \beta = 2$

$$y = e^{-x/2}(C_1 \cos 2x + C_2 \sin 2x)$$

4 Use $y(0) = -1$ first

$$-1 = e^0(C_1 \cos 0 + C_2 \sin 0) = C_1$$

$$\boxed{C_1 = -1}$$

At zero: $e^0 = 1$, $\cos 0 = 1$, and $\sin 0 = 0$. That's why the whole equation collapses to just C_1.

EXAMPLE 04 / THE MISSING WORK

Differentiate with the product rule

This is the part your professor described but did not fully write out.

GENERAL SOLUTION

$$y = \underbrace{e^{-x/2}}_{f(x)}\underbrace{(C_1\cos 2x + C_2\sin 2x)}_{g(x)}$$

The rule is $\boxed{(fg)' = f'g + fg'}$

A **Differentiate the first factor**

$$f' = (-\tfrac{1}{2})e^{-x/2}$$

B **Differentiate the second factor**

$$\begin{aligned} g' &= C_1(-2\sin 2x) + C_2(2\cos 2x) \\ &= -2C_1\sin 2x + 2C_2\cos 2x \end{aligned}$$

C **Plug both pieces into the product rule**

$$\begin{aligned} y' = &-\tfrac{1}{2}e^{-x/2}(C_1\cos 2x + C_2\sin 2x) \\ &+ e^{-x/2}(-2C_1\sin 2x + 2C_2\cos 2x) \end{aligned}$$

Now evaluate at $x = 0$. All $\sin 0$ terms vanish. $\cos 0 = 1$:

$$y'(0) = -\tfrac{1}{2}C_1 + 2C_2$$

You do not have to fully expand y' before substituting zero. Doing it in two separate product-rule pieces often avoids mistakes.

EXAMPLE 04 / FINISH IT

Find C_2 and check the answer

This is how the professor got the $\frac{3}{4}$ in your notes.

You already know: $y'(0) = 2, \qquad C_1 = -1, \qquad y'(0) = -\frac{1}{2}C_1 + 2C_2$

1 Substitute

$$2 = -\tfrac{1}{2}(-1) + 2C_2$$

2 Multiply the negatives

$$2 = \tfrac{1}{2} + 2C_2$$

3 Move $\frac{1}{2}$ to the left

$$2 - \tfrac{1}{2} = 2C_2$$

$$\tfrac{3}{2} = 2C_2$$

4 Divide both sides by 2

$$C_2 = \frac{3/2}{2} = \frac{3}{4}$$

FINAL ANSWER

$$\boxed{y = e^{-x/2}\left(-\cos 2x + \tfrac{3}{4}\sin 2x\right)}$$

Quick verification: $y(0) = -1$. Also, $y'(0) = \frac{1}{2} + 2(\frac{3}{4}) = \frac{1}{2} + \frac{3}{2} = 2$. Both initial values match.

07 / MOVE TO HIGHER ORDER

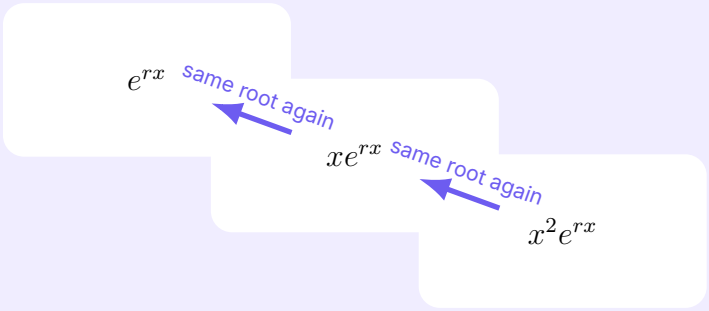
Third- and fourth-order equations

The solving method stays the same. There are just more roots to handle.

POWER MATCHING ALWAYS FOLLOWS THE DERIVATIVE ORDER

y''''	y'''	y''	y'	y
m^4	m^3	m^2	m	1

Repeated-root staircase



Repeated root of multiplicity k means it appears k times. Use:

$$e^{rx},\ xe^{rx},\ x^2e^{rx},\ \ldots,\ x^{k-1}e^{rx}$$

Each root contributes one independent term in the answer.

Missing derivatives mean missing powers! For $y''' + 3y'' - 4y = 0$, there is no y', so the auxiliary equation is $m^3 + 3m^2 + \underbrace{0m}_{\text{missing}} - 4 = 0$.

EXAMPLE 05 / THIRD ORDER

One new root and a repeated pair

The same three-case logic works with a cubic.

QUESTION

$$\boxed{y''' + 3y'' - 4y = 0}$$

1 Auxiliary equation (no m term)

$$m^3 + 3m^2 - 4 = 0$$

2 Try $m = 1$, one of the factors of 4

$$1^3 + 3(1)^2 - 4 = 1 + 3 - 4 = 0$$

So $(m - 1)$ divides the polynomial.

3 Synthetic division (include the $0m$ slot)

1	1	3	0	−4
		1	4	4
	1	4	4	0

The remainder is 0. The bottom row becomes $m^2 + 4m + 4$.

4 Factor what remains

$$(m - 1)(m^2 + 4m + 4) = (m - 1)(m + 2)^2$$

$$\boxed{m = 1,\ -2,\ -2}$$

FINAL ANSWER

$$\boxed{y = C_1 e^x + C_2 e^{-2x} + C_3 x e^{-2x}}$$

You are combining Case 1 and Case 2. 1 produces e^x. The two copies of -2 produce e^{-2x} and xe^{-2x}.

EXAMPLE 06 / FOURTH ORDER

Repeated complex roots

The final problem combines the repeated-root rule with Euler's sine/cosine rule.

QUESTION

$$\boxed{y'''' + 2y'' + y = 0}$$

1 Auxiliary polynomial

$$m^4 + 2m^2 + 1 = 0$$

2 Recognize the perfect square

$$(m^2 + 1)^2 = 0$$

3 Find all four roots, counting repeats

$$m^2 = -1 \quad \Rightarrow \quad m = \pm i$$

$$\boxed{m = i,\ i,\ -i,\ -i}$$

4 Complex pair gives cosine and sine

$$\cos x, \qquad \sin x$$

5 The pair repeats, so multiply both by x

$$x\cos x, \qquad x\sin x$$

FINAL ANSWER

$$\boxed{y = C_1\cos x + C_2\sin x + x(C_3\cos x + C_4\sin x)}$$

Where did the exponential go? For roots $\pm i$, the real part is $\alpha = 0$. Therefore $e^{\alpha x} = e^{0x} = 1$, so it disappears.

08 / THE EXAM PLAYBOOK

How to solve ANY problem here

A single repeatable approach for all six class questions.

1. **Recognize:** linear + homogeneous + constant coefficients?
2. **Convert:** $y \to 1,\ y' \to m,\ y'' \to m^2$, etc.
3. **Solve:** factor or use the quadratic formula.
4. **Classify:** different, repeated, or complex roots.
5. **Write:** one independent function per root.
6. **Finish:** if initial conditions are given, find constants.

Formula wall: copy this into your quick-revision notes.

Distinct: $C_1 e^{m_1 x} + C_2 e^{m_2 x}$

Repeated: $(C_1 + C_2 x)e^{mx}$

Complex: $e^{\alpha x}(C_1 \cos \beta x + C_2 \sin \beta x)$

Triple repeat: $(C_1 + C_2 x + C_3 x^2)e^{mx}$

Top mistakes to avoid: dropping the extra x, forgetting a missing $0m$ in synthetic division, treating i as part of β, or differentiating only one factor during an IVP.

09 / CHECK YOUR UNDERSTANDING

Three quick practice prompts

Try them before looking at the small answers at the bottom.

01 **Find the root case and general solution**

$$y'' - 3y' + 2y = 0$$

02 **Find the root case and general solution**

$$y'' + 6y' + 9y = 0$$

03 **Find the root case and general solution**

$$y'' + 2y' + 5y = 0$$

Answers

1. $(m-1)(m-2) = 0$. Distinct, so $y = C_1 e^{x} + C_2 e^{2x}$.

2. $(m+3)^2 = 0$. Repeated, so $y = (C_1 + C_2 x)e^{-3x}$.

3. $m = -1 \pm 2i$. Complex, so $y = e^{-x}(C_1 \cos 2x + C_2 \sin 2x)$.

What matters most for your next tutorial: Don't memorize example numbers. Memorize the conversion $y^{(n)} \mapsto m^n$, know the three formulas, and practice finding roots accurately.

End of illustrated lesson. First 3 pages: your complete original handwriting. Subsequent pages: explained examples derived from the supplied October 21 transcript and notes.